(iii) $\int \frac{1}{(3x-2)}dx = \frac{1}{3}\log|(3x-2)|+c$ (iv) $\int \frac{1}{(5-4x)}dx = -\frac{1}{4}\log|(5-4x)|+c$

Example 11.7

Integrate the following with respect to x:

(i) $\frac{1}{1+(2x)^2}$ (ii) $\frac{1}{\sqrt{1-(9x)^2}}$ (iii) $\frac{1}{\sqrt{1-25x^2}}$

Solution

(i) $\int \frac{1}{1+(2x)^2}dx = \frac{1}{2}\tan^{-1}(2x)+c$ (ii) $\int \frac{1}{\sqrt{1-(9x)^2}}dx = \frac{1}{9}\sin^{-1}(9x)+c$

(iii) $\int \frac{1}{\sqrt{1-25x^2}}dx = \int \frac{1}{\sqrt{1-(5x)^2}}dx = \frac{1}{5}\sin^{-1}(5x)+c$

EXERCISE 11.2

Integrate the following functions with respect to x:

(1) (i) $(x+5)^6$ (ii) $\frac{1}{(2-3x)^4}$ (iii) $\sqrt{3x+2}$

(2) (i) $\sin 3x$ (ii) $\cos(5-11x)$ (iii) $\text{cosec}^2(5x-7)$

(3) (i) e^{3x-6} (ii) e^{8-7x} (iii) $\frac{1}{6-4x}$

(4) (i) $\sec^2\frac{x}{5}$ (ii) $\text{cosec}(5x+3)\cot(5x+3)$ (iii) $\sec(2-15x)\tan(2-15x)$

(5) (i) $\frac{1}{\sqrt{1-(4x)^2}}$ (ii) $\frac{1}{\sqrt{1-81x^2}}$ (iii) $\frac{1}{1+36x^2}$

11.5 Properties of Integrals

(1) If k is any constant, then $\int kf(x)dx = k\int f(x)dx$

(2) $\int (f_1(x) \pm f_2(x))dx = \int f_1(x)dx \pm \int f_2(x)dx$

Note 11.1

The above two properties can be combined and extended as

$$\int (k_1 f_1(x) \pm k_2 f_2(x) \pm k_3 f_3(x) \pm \cdots \pm k_n f_n(x))\,dx$$

$$= k_1\int f_1(x)dx \pm k_2\int f_2(x)dx \pm k_3\int f_3(x)dx \pm \cdots \pm k_n\int f_n(x)dx.$$

That is, the integration of the linear combination of a finite number of functions is equal to the linear combination of their integrals

Example 11.8

Integrate the following with respect to x:

(i) $5x^4$ (ii) $5x^2 - 4 + \frac{7}{x} + \frac{2}{\sqrt{x}}$ (iii) $2\cos x - 4\sin x + 5\sec^2 x + \text{cosec}^2 x$

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